

## Q-1: Maze of Amaravati

In the ancient Maze of Amaravati, a notorious labyrinth in India, every room holds a deadly secret. The maze is arranged as an  $n \times m$  grid, and each room is guarded by a time threshold. `moveTime[i][j]` represents the minimum time in seconds when you can start moving to that room. If you start making a move to a room before its safe time (specified in the array `moveTime`), you will die.

You start at the entrance, room  $(0, 0)$ , at time  $t = 0$ . From any room, you can move to an adjacent one (rooms that share a common wall), but beware: your moves alternate in duration. The first move takes 1 second, the second takes 2 seconds, then 1 second again, and so on.

Your goal is to determine the minimum time needed to safely reach the exit at room  $(n - 1, m - 1)$ .

### Example Test Cases:

#### TEST CASE 1

Input:

`moveTime = [[0,4],[4,4]]`

Output:

7

Explanation:

At time  $t = 4$ , move from room  $(0, 0)$  to room  $(1, 0)$  in 1 second. You cannot start making a move at  $t = 3$  and reach at  $t = 4$ . You need to make a move at  $t = 4$ , thus reaching at  $t = 5$ .

At time  $t = 5$ , move from room  $(1, 0)$  to room  $(1, 1)$  in 2 seconds.

**TEST CASE 2**

Input:

`moveTime = [[0,0,0,0],[0,0,0,0]]`

Output:

6

Explanation:

At time  $t = 0$ , move from room  $(0, 0)$  to room  $(1, 0)$  in 1 second.At time  $t = 1$ , move from room  $(1, 0)$  to room  $(1, 1)$  in 2 seconds.At time  $t = 3$ , move from room  $(1, 1)$  to room  $(1, 2)$  in 1 second.At time  $t = 4$ , move from room  $(1, 2)$  to room  $(1, 3)$  in 2 seconds.**TEST CASE 3**

Input:

`moveTime = [[0,1],[1,2]]`

Output:

4

**Input Details:**

- The maze consists of  $n$  rows and  $m$  columns. Rooms are indexed from  $(0, 0)$  (entrance) to  $(n - 1, m - 1)$  (exit).
- The 2D array `moveTime` of size  $n \times m$  provides the earliest time at which you can safely enter each room.
- Moving between adjacent rooms takes an alternating duration: 1 second for the first move, 2 seconds for the second move, then 1 second, and so on.
- Entering a room before its designated time results in instant death.

**Output Expectations:**

- Print a single integer: the minimum time required to reach the exit safely.

**Constraints:**

- $2 \leq n \leq 750$  (number of rows)

- $2 \leq m \leq 750$  (number of columns)
- $0 \leq \text{moveTime}[i][j] \leq 10^9$

## Q-2: The Indian Railway Challenge

Raj and Priya, two friends from India, are planning a train journey from Mumbai to Varanasi. To ensure a smooth experience, they decide to book their tickets through the official online railway booking platform, known for its convenience and efficiency.

To make their trip efficient, they seek answers to the following:

1. What is the minimum fare required to travel from Mumbai to Varanasi?
2. How many distinct routes offer this minimum fare? (Provide the answer modulo  $10^9 + 7$ )
3. Among these cost-effective routes, what is the shortest sequence of train journeys?
4. Conversely, what is the longest sequence of train journeys within these minimum-fare pathways?

### Example Scenario:

**TEST CASE 1**

Input:

```
4 5
1 4 5
1 2 4
2 4 5
1 3 2
3 4 3
```

Output:

```
5 2 1 2
```

### Input Details:

- The journey encompasses  $n$  cities and  $m$  one-way train routes connecting them. Cities are labeled sequentially from 1 to  $n$ , with Mumbai designated as City 1 and Varanasi as City  $n$ .
- Each subsequent line provides information about a train route, detailing:
  - $a$ : Departure city
  - $b$ : Arrival city
  - $c$ : Fare associated with this train journey
- It is guaranteed that at least one viable route exists from Mumbai to Varanasi.

**Output Expectations:**

- Four integers should be printed in sequence, corresponding to the queries posed above.

**Operational Constraints:**

- $1 \leq n \leq 100,000$  (Number of cities)
- $1 \leq m \leq 200,000$  (Number of train routes)
- $1 \leq a, b \leq n$  (City indices)
- $1 \leq c \leq 1,000,000,000$  (Train fare)

**Note:** Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.

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